

IT-601

Examination –April.- 2025

B.Tech. VI Sem : Information Technology

Subject name: Data Mining and Warehousing

Time : 3 Hrs

Max. Marks : 60

Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part (a)- Objective type/Fill in the blanks/True & False

Part (b) – Three/four sentences based answer question (Maximum 50 words)

Part (c) – Short Descriptive Ans./Numerical/ algorithm /programs based Question.
(Maximum 100 words)

Part (d) – Long Answer /Shorts -notes /Descriptive Ans. based Question (Maximum 400 words)

Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	(i) _____ is a subject-oriented, integrated, time-variant, nonvolatile collection of data in support of management decisions. A. Data Mining B. Data Warehousing C. Web Mining D. Text Mining (ii) The data is stored, retrieved & updated in _____. A. OLAP B. OLTP C. ROLAP D. MOLAP	02	1	1
	(b)	What is Data Mart. What is the role of Data Mart in data warehousing?	02	4	1
	(c)	Differentiate between OLAP and OLTP?	03	3	1
OR					
		Describe various Data Cube Operations?	03	2	1
	(d)	What do you mean by data warehouse modeling? Explain data warehouse schemas for multidimensional data models?	05	4	1
OR					
		Define data warehouse with five features. With neat sketch explain the architecture of data warehouse with suitable block diagram.	05	3	1
Q.2	(a)	(i) Which is the correct process of data mining? a) Infrastructure, exploration, analysis, interpretation and exploitation b) Exploration, Infrastructure, analysis, interpretation and exploitation c) Exploration, Infrastructure, interpretation, analysis and exploitation d) Exploration, Infrastructure, analysis, exploitation and analysis (ii) Which of the following statement about knowledge and data discovery management system (KDDMS) is false? a) It will provide concurrency features	02	1	2

		b) It will provide recovery features c) It will include data mining tools and data management tools d) It will include data mining tools but not data management tools													
	(b)	What are the different tasks of Data Mining?	02	4	2										
	(c)	Explain the process of KDD?	03	3	2										
OR															
		Define the following : i. Data cleaning ii. Data transformation	03	2	2										
	(d)	Write a difference between classification and clustering?	05	4	2										
OR															
		Describe the applications and challenges of Data Mining?	05	2	2										
Q.3	(a)	(i) What is the relation between a candidate and frequent itemsets? a) A candidate itemset is always a frequent itemset b) A frequent itemset must be a candidate itemset c) No relation between these two d) Strong relation with transactions ii) Which algorithm requires fewer scans of data? (a) Apriori (b) FP Growth (c) Naive Bayes (d) Decision Trees	02	1	3										
	(b)	Describe Association Rule?	02	4	3										
	(c)	Apply Apriori algorithm and find all the frequent item sets for the transaction given in the table: Min Sup=50% <table><tr><th>TID</th><th>Items Bought</th></tr><tr><td>100</td><td>1, 3, 4</td></tr><tr><td>200</td><td>2, 3, 5</td></tr><tr><td>300</td><td>1, 2, 3, 5</td></tr><tr><td>400</td><td>2, 5</td></tr></table>	TID	Items Bought	100	1, 3, 4	200	2, 3, 5	300	1, 2, 3, 5	400	2, 5	03	3	3
TID	Items Bought														
100	1, 3, 4														
200	2, 3, 5														
300	1, 2, 3, 5														
400	2, 5														
OR															
		What do you mean by Market Basket analysis and how it can help in a supermarket?	03	2	3										
	(d)	Find the frequent itemsets for the transactional data table,given below ,without candidate generation. Use the FP-Tree Growth algorithm approach. Min_Sup=2 <table><tr><th>TID</th><th>List of Item_IDs</th></tr><tr><td>T100</td><td>11,I2,I5</td></tr><tr><td>T200</td><td>12,I4</td></tr></table>	TID	List of Item_IDs	T100	11,I2,I5	T200	12,I4	05	1	3				
TID	List of Item_IDs														
T100	11,I2,I5														
T200	12,I4														

		T300	I2,I3			
		T400	I1,I2,I4			
		T500	I1,I3			
		T600	I2,I3			
		T700	I1,I3			
		T800	I1,I2,I3,I5			
		T900	I1,I2,I3			
OR						
		Explain Performance Parameter of Algorithms?	05	4	3	
Q.4	(a)	(i) Which of the following clustering requires merging approach? a) Partitional b) Hierarchical c) Naive Bayes d) None of the mentioned (ii) K-means is not deterministic and it also consists of number of iterations. a) True b) False	02	2	4	
	(b)	What is Cluster analysis?	02	1	4	
	(c)	What do you mean by Overfitting?	03	2	4	
OR						
		What is Gini Index? How do you calculate Gini index from data?	03	2	4	
	(d)	What are the three phases of construction of the decision tree? Describe the importance of each of the phases?	05	3	4	
OR						
		What are the different phases of BIRCH? How are they important in clustering?	05	4	4	
Q.5	(a)	(i) Web mining - is the application of _____ a) Data Mining b) Text Mining c) Both a and b d) None of these (ii) Web structure mining is the process of discovering _____ information from the web a) Semi structured b) Unstructured c) Structured d) None of the above	02	1	5	
	(b)	What is Temporal Mining?	02	2	5	
	(c)	Define the Ethical Issues?	03	3	5	
OR						
		Explain Multimedia data mining with suitable example?	03	3	5	

	(d)	What is Spatial data mining? how do you handle spatial and non spatial data, while carrying out any mining task?	05	4	5
OR					
		What is Web mining? How is web usage mining different from web structure mining and web content mining?	05	4	5
Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating					
